

Benjamin F Randall

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Professional Summary

Computer Engineering student focused on **embedded systems, firmware, and real-time control**. Experienced in RTOS-based microcontroller development, digital logic, and hardware-software integration. Leadership in IEEE and IDEA Club demonstrates initiative, collaboration, and communication with technical and non-technical stakeholders. Seeking internship or junior engineering roles developing and testing embedded hardware and real-time applications.

Skills

- **Technical Skills:** SQL, C, MATLAB, SolidWorks, Linux OS, HTML
- **Engineering Tools:** Oscilloscopes, Multimeters, Logic Analyzers
- **Soft Skills:** Problem-solving, teamwork, communication, adaptability

Work Experience

Diesel Direct, IT intern

June 2021 – Present

- Handled incoming caller problems efficiently, ensuring minimal downtime for users and systems.
- Assisted in expanding and organizing the company knowledge base to improve issue resolution time.
- Organized, cataloged, and updated hardware and software inventory to support smoother technology deployment.

Education

University of Massachusetts Dartmouth, Dartmouth, MA

Expected Graduation: May 2026

Bachelor of Science in Computer Engineering, Concentration in Cybersecurity

- Relevant coursework: Embedded Systems Design, Digital Logic Design, Circuit Analysis, and Computer Architecture
- Member of IEEE Student Branch, actively participating in networking events and technical workshops.

Selected Projects

FPGA Digital Clock with Alarm – Nexys A7 (Board level VHDL)

- Designed HH:MM:SS timekeeping with AM/PM indicator and alarm trigger on SSD display.
- Developed debounced button control for set/reset operations and alarm logic.
- Synthesized in Vivado and verified timing using on-board clock resources.

Solar Tracker System – ATmega Microcontroller & Servo Control

- Implemented light-angle tracking using photoresistor array and PID loop for servo motor actuation.
- Optimized pulse width to maintain alignment within 2° of peak illumination.

Networking Sensor Node – ESP32 and ESP8266 Wi-Fi AP/Client Design

- Configured DHCP-based communication between ESP devices with TCP/UDP handshake.
- Implemented server-client model to transmit sensor data and acknowledgments in real time..

References Available upon request